

Assignment1: CancerMortality

Joan Acero, Anna Monsó, Walter Troiani

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HINT: Retain all available numeric variables except those that are not useful in explaining the target. # Questions:	

- Should we remove the outliers, apply transformations, etc.in the first analysis of the variables (Variable Analysis) ?

Load the dataset and required libraries

```
rm(list=ls())
knitr::opts_chunk$set(fig.align='center', out.width='50%') # Manage output plots width and align

train <- read.csv("train.csv")
test  <- read.csv("test.csv")

library(moments)
library(car)
```

```
## Loading required package: carData
```

Define some functions to avoid repetition of same processes

```
calculate_outliers <- function(data) {
  # Check for regular outliers
  outliers <- boxplot.stats(data)$out
  num_outliers <- length(outliers)

  # Calculate the lower and upper bounds for severe outliers
  Q1 <- quantile(data, 0.25)
  Q3 <- quantile(data, 0.75)
  IQR <- Q3 - Q1

  lower_severe_bound <- Q1 - 3 * IQR
  upper_severe_bound <- Q3 + 3 * IQR

  # Identify severe outliers
  severe_outliers <- data[data < lower_severe_bound | data > upper_severe_bound]
  num_severe_outliers <- length(severe_outliers)

  # Return the results as a list
  return(list(
    num_outliers = num_outliers,
    num_severe_outliers = num_severe_outliers
  ))
}
```

Data Preparation

In this section, a descriptive analysis is conducted on each variable, a data quality report is generated, and imputation and profiling are addressed.

Univariate Descriptive Analysis (Variable Analysis)

- Original numeric variables corresponding to qualitative concepts have to be converted to factors:
- Original numeric variables corresponding to real quantitative concepts are kept as numeric but additional factors should also be created
- Exploratory Data Analysis for each variable (numeric summary and graphic support) of the train sample.

Variable *avganncount*

avganncount is a numerical variable representing the “Mean number of reported cancer cases diagnosed annually (2010-2015).” The difference between the mean (623.2) and the median (175) indicates that the distribution is not symmetric and deviates from normality. Additionally, the first and third quartiles (80 and 509, respectively) are much lower than the mean of 623.2 and the maximum value of 38150, further suggesting skewness. The boxplot confirms this skewness, with an extended upper whisker and numerous outliers, indicating a heavy right tail. It has 273 outliers, and 224 of these are severe outliers.

This implies that while most annual cancer case counts are relatively low, a few instances are exceptionally high, likely due to large population centers or higher cancer incidence rates. The histogram overlaid with a normal curve highlights the deviation from a normal distribution with the same mean and standard deviation. This variable has no missing values. Lastly, the skewness indicator and shapiro test rejection of the null hypothesis also confirm the non-normal distribution of the variable.

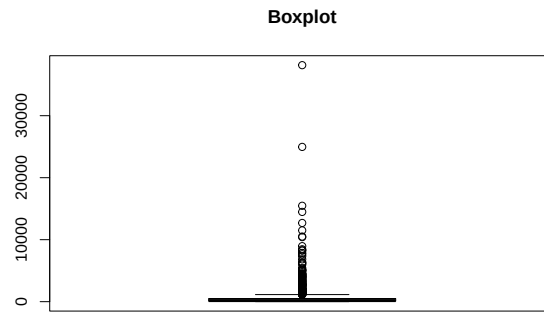
```
summary <- summary(train$avganncount); summary
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       7.0    80.0   175.0   623.2   509.0 38150.0
```

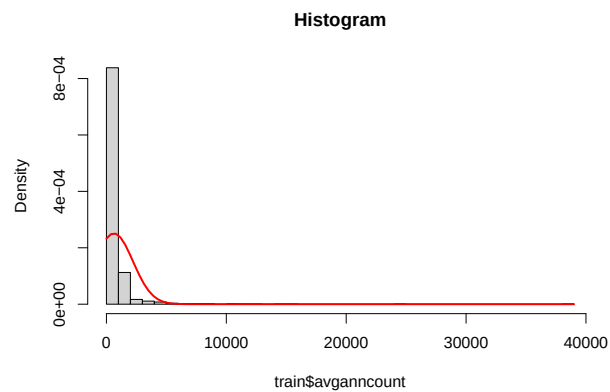
```
mm <- mean(train$avganncount)
ss <- sd(train$avganncount)
range <- (summary[[6]] - summary[[1]]); range
```

```
## [1] 38143
```

```
boxplot(train$avganncount, main= "Boxplot")
```



```
hist(train$avganncount,freq = F,30, main = "Histogram")
curve(dnorm(x, mean = mm, sd = ss),col="red", lwd = 2, add=T)
```



```
skewness(train$avganncount)
```

```
## [1] 11.48879
```

```
shapiro.test(train$avganncount)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  train$avganncount
## W = 0.33377, p-value < 2.2e-16
```

```
calculate_outliers(train$avganncount)
```

```
## $num_outliers
## [1] 273
##
## $num_severe_outliers
## [1] 224
```

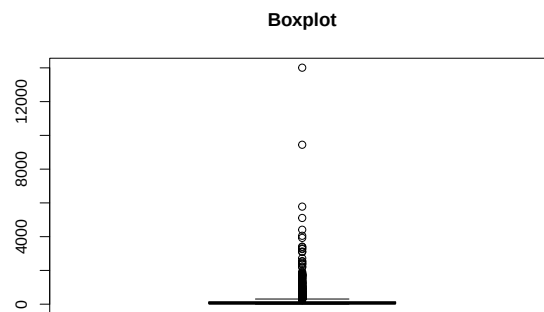
Variable *avgdeathsperyear*

In a similar way, *avgdeathsperyear* is a numerical variable expressing the “Mean number of reported mortalities due to cancer”. It has no missing values. The difference between the mean (191.6) and median (62) indicates that the distribution is not symmetric. Moreover, the first and third quartiles (29 and 140.5, respectively) are lower than the mean and the maximum value of 14010, further suggesting a skewed distribution. With the help of graphical tools like the boxplot we can confirm the skeweness, seeing 225 outliers (138 of which are severe outliers), indicating a large right tail. The histogram overlaid with a normal curve highlights the deviation from a normal distribution with the same mean and standard deviation, and the Shapiro test also confirms the non-normality of the variable.

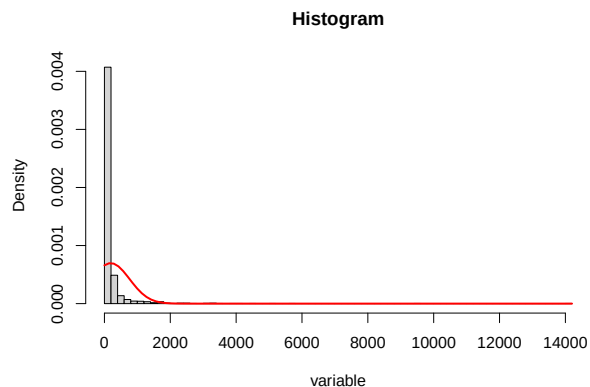
```
variable <- train$avgdeathsperyear;
summary <- summary(variable)
mm <- mean(variable)
ss <- sd(variable)
range <- (summary[[6]] - summary[[1]]); range
```

```
## [1] 14007
```

```
boxplot(variable, main = "Boxplot")
```



```
hist(variable, freq = F, 100, main = "Histogram")
curve(dnorm(x, mean = mm, sd = ss), col = "red", lwd = 2, add = T)
```



```
skewness(variable)
```

```
## [1] 12.4999
```

```
shapiro.test(variable)
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: variable  
## W = 0.26769, p-value < 2.2e-16
```

```
calculate_outliers(variable)
```

```
## $num_outliers  
## [1] 225  
##  
## $num_severe_outliers  
## [1] 138
```

Variable *target_deathrate* :

Variable *target_deathrate* is the target of the model. It contains no missing values in the train sample. The median (178.3) and mean (178.8) values are really close, expressing the symmetry of the distribution of the variable. Thanks to the boxplot and the histogram we can graphically confirm the symmetry of the distribution.

We performed the Shapiro test to check for the normality of the data, and obtained a small p-value that rejects the null hypothesis, indicating that the variable does not follow a normal distribution. Therefore, the target variable is not normally distributed.

Moreover, it contains 35 outliers, 1 of which is a severe outlier.

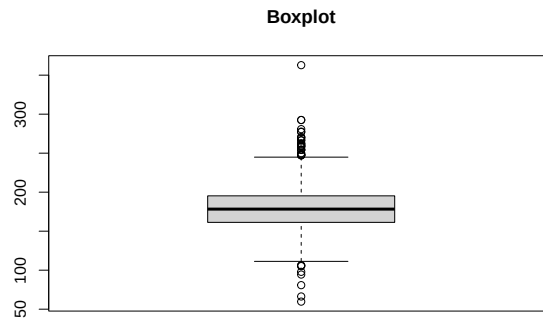
```
variable <- train$target_deathrate;  
summary <- summary(variable); summary
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##      59.7   161.3   178.3   178.8   195.3   362.8
```

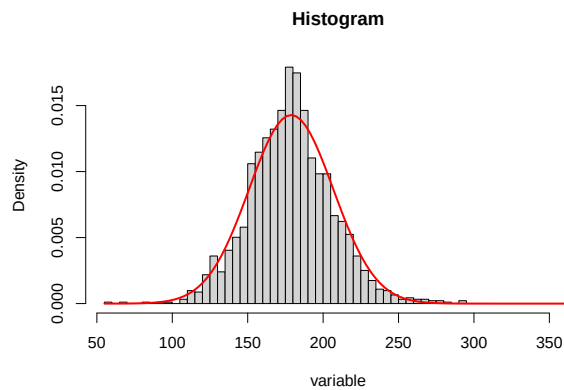
```
mm <- mean(variable)  
ss <- sd(variable)  
range <- (summary[[6]] - summary[[1]]); range
```

```
## [1] 303.1
```

```
boxplot(variable, main = "Boxplot")
```



```
hist(variable,freq = F,100, main = "Histogram")
curve(dnorm(x, mean = mm, sd = ss),col="red", lwd = 2, add=T)
```



```
skewness(variable)
```

```
## [1] 0.3243216
```

```
shapiro.test(variable)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  variable
## W = 0.98647, p-value = 4.149e-12
```

```
calculate_outliers(variable)
```

```
## $num_outliers
## [1] 35
##
## $num_severe_outliers
## [1] 1
```

Variable *incidencerate* :

Variable *incidencerate* expresses “Mean per capita (100,000) cancer diagnoses”. The variable has no missing values. It has similar mean (449) and median (453.5) values given its range. It contains 60 outliers, 7 of which are severe outliers. In the boxplot we can see how the most severe outliers are high and not low values, corresponding to Osceola County, Florida and Williamsburg city, Virginia. The histogram with the overlapping normal distribution shows a quite symmetrical distribution, but the Shapiro test and skeweness value discard the normality of the variable’s distribution.

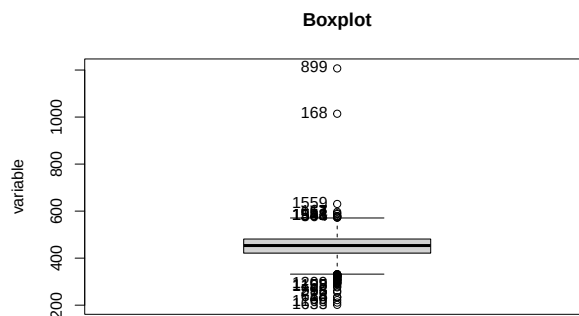
```
variable <- train$incidencerate;
summary <- summary(variable); summary
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      201.3  421.4   453.5   449.0  481.3  1206.9
```

```
mm <- mean(variable)
ss <- sd(variable)
range <- (summary[[6]] - summary[[1]]); range
```

```
## [1] 1005.6
```

```
Boxplot(variable, main = "Boxplot")
```



```
## [1] 1633 1168 60 18 634 295 558 1122 1155 1209 899 168 1559 167 17
## [16] 954 1558 1548 1541 364
```

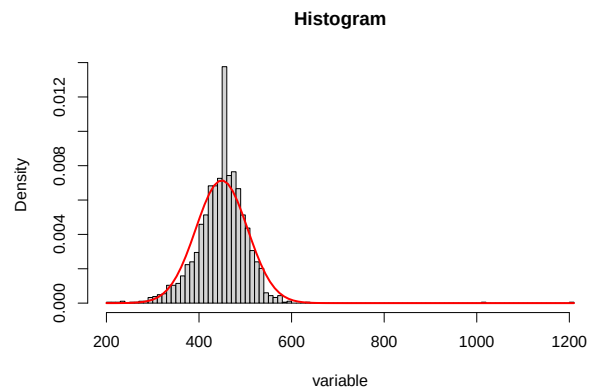
```
train[889, "geography"]
```

```
## [1] "Osceola County, Florida"
```

```
train[168, "geography"]
```

```
## [1] "Williamsburg city, Virginia"
```

```
hist(variable,freq = F,100, main = "Histogram")
curve(dnorm(x, mean = mm, sd = ss),col="red", lwd = 2, add=T)
```



```
skewness(variable)
```

```
## [1] 1.429279
```

```
shapiro.test(variable)
```

```
##
## Shapiro-Wilk normality test
##
## data: variable
## W = 0.89577, p-value < 2.2e-16
```

```
calculate_outliers(variable)
```

```
## $num_outliers
## [1] 60
##
## $num_severe_outliers
## [1] 7
```

Data Quality Report

Per variable, count:

- Number of missing values.
- Number of errors (including inconsistencies).
- Number of outliers. -Rank variables according the sum of missing values (and errors).

```
# Number of missing values.
# Number of errors (including inconsistencies).
# Number of outliers.
# Rank variables according the sum of missing values (and errors).
```

Per individuals, count:

- Number of missing values
- Number of errors
- Number of outliers
- Identify individuals considered as multivariant outliers.

```
#Number of missing values
#Number of errors
#Number of outliers
#Identify individuals considered as multivariant outliers.
```

Create a variable adding the total number of missing values, outliers and errors. Describe these variables, to which other variables exist higher associations.

- Compute the correlation with all other variables. Rank these variables according the correlation
- Compute for every group of individuals the mean of missing/outliers/error values. Rank those groups according the computed mean.

```
#Include Code
```

Imputation

- Numeric Variables
- Factors

```
#Include Code
```

Profiling

- Target : Death rate

Model fitting to train sample only!!

```
#Include Code
```